Polynomial total mistake bounds for mean-field Bayesian tanh networks

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Abstract

We study shallow Bayesian tanh networks with independent Gaussian weights at mean-field scaling. Training conditions on a signed output margin greater than $1/2$ at each observed input. The initial prior has no output restriction at unobserved inputs. If a Boolean target on d bits has a width-C teacher with bounded weights and margin greater than $1/2$, then, at every sufficiently large learner width, the total number of online confidence mistakes is $O(d^4 + C^4)$, with the weight bound and positive prior prefactor fixed. The input order may be arbitrary or adaptive. A confidence mistake means that the posterior assigns at most a fixed probability, such as $2/3$, to the correct sign; any fixed test margin below $1/2$ is also allowed. Fresh posterior sampling satisfies corresponding expected and high-probability total error bounds. The proof identifies the large-width posterior with successive relative-entropy projections and shows that each bad prediction consumes a fixed amount of a bounded entropy budget. We give an explicit Gaussian mixture certificate for that budget. The width threshold is not quantified, and the result is not uniform as the prior prefactor tends to zero. The absence of an initial global output-margin restriction is essential to the convex projection argument.

1 Model and result

A small target can admit a polynomial IID sample bound without admitting a polynomial bound on total mistakes along an arbitrary input sequence. Here we prove the stronger online conclusion for a Bayesian neural-network model that requires a positive output margin on observed examples, but imposes no margin restriction at unobserved inputs before learning starts.

Fix $d \geq 1$, let $X = \{-1, 1\}^d$, and write $D = d + 1$, $M = 2^d$, and $\bar{x} = (1, x) \in \mathbb{R}^D$. Fix $\tau > 0$. At hidden width N, sample independently

$$u_j \sim \mathcal{N}(0, \tau^2 I_D/d), \qquad a_j \sim \mathcal{N}(0, \tau^2), \qquad 1 \leq j \leq N, \tag{1}$$

and define

$$F_N(x) = \frac{1}{N} \sum_{j=1}^{N} a_j \tanh(u_j \cdot \bar{x}). \tag{2}$$

The physical readout weights $v_j = a_j/N$ have variance τ^2/N^2. These are the canonical shallow mean-field variances. Equivalently, in the normalization with first-layer variance $N\sigma_N^2/d$ and readout variance σ_N^2/N, they correspond to $\sigma_N = \tau/\sqrt{N}$. We take $N \to \infty$ at fixed τ, not a cold limit at fixed width.

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Let $f : X \to \{-1, 1\}$ be the target. A history is a subset $S \subseteq X$ whose labels have been revealed. The posterior is precisely the prior conditioned on

$$f(x)F_N(x) > 1/2 \qquad (x \in S). \tag{3}$$

There is *no initial conditioning* on the outputs elsewhere. For a test-margin threshold $0 \le h < 1/2$, define

$$p_{N,S,h}(x) = \mathbb{P}\{f(x)F_N(x) > h \mid (3)\}.$$

Fix $0 < \theta < 1$. An online confidence mistake means $p_{N,S,h}(x) \le \theta$. Examples are $h = 0$, which tests the sign, and $h = 1/4$, which requires a smaller margin than training. We may take $\theta = 2/3$ in either case.

After a prediction, the true label is revealed and the input joins S. The next input may be chosen adaptively. Repeated observed inputs cause no confidence or sampling errors: their posterior signed output is already greater than $1/2$. Let ρ_0 denote the one-neuron law in (1), and use $\mathrm{KL}(P\|Q)$ for relative entropy with natural logarithms.

Theorem 1 (Total online mistakes)**.** *Suppose a probability law ρ_T on $(u, a) \in \mathbb{R}^D \times \mathbb{R}$ satisfies*

$$\mathrm{KL}(\rho_T\|\rho_0) \le K_T < \infty, \qquad f(x)\mathbb{E}_{\rho_T}[a \tanh(u \cdot \bar{x})] > 1/2 \quad (x \in X). \tag{4}$$

Set $c_0 = (\log 2)/2$ and

$$R_h = \frac{16\tau^2 K_T(K_T + c_0)}{(1/2 - h)^2}. \tag{5}$$

For each fixed $0 < \theta < 1$ and $0 \le h < 1/2$, there exists N_0 such that every $N \ge N_0$ gives at most $\lfloor R_h \rfloor$ total confidence mistakes. The same N_0 works for every input order, including adaptive choices of the next input.

If predictions use the sign of a fresh posterior sample at each round, then for every $\eta > 0$ there exists N_1 such that every $N \ge N_1$ satisfies

$$\mathbb{E}[\textit{total sign errors}] \le R_0 + \eta, \tag{6}$$

$$\mathbb{P}\{\textit{total sign errors} \le \lfloor R_0 \rfloor\} \ge 1 - \eta. \tag{7}$$

These statements are uniform over adaptive input orders as well.

The width thresholds may depend on the target and all fixed parameters, and need not be polynomial. The conclusion concerns every sufficiently large width, rather than every width capable of representing the teacher. The confidence threshold affects the width threshold but does not affect the displayed mistake bound.

Corollary 2 (A bounded teacher gives a polynomial bound)**.** *Suppose*

$$F_T(x) = \sum_{i=1}^{C} v_i \tanh(u_i \cdot \bar{x}), \qquad f(x)F_T(x) > 1/2 \quad (x \in X),$$

where $C \ge 1$ and all coordinates of u_i and all v_i have absolute value at most $B > 0$. Put

$$s = \min\left\{\frac{\tau}{\sqrt{d}}, \frac{1}{8CB\sqrt{D}}\right\}, \qquad q = \frac{ds^2}{\tau^2} \in (0, 1], \tag{8}$$

and

$$K_T = \frac{dDB^2 + 4C^2B^2}{2\tau^2} + \frac{D}{2}(q - 1 - \log q). \tag{9}$$

Then Theorem 1 applies with this value of K_T. At fixed B, τ, h, the total confidence mistake bound is $O(d^4 + C^4)$. The same polynomial controls the expected and high-probability sampling bounds for the sign convention.

The result is not uniform as $\tau \to 0$. Even (9) has an inverse-τ^2 contribution. The theorem does not supply a total mistake bound for a subsequent zero-temperature limit.

2 The large-width posterior is an entropy projection

For one neuron, define its full output vector and its feature vector by

$$Z(u, a) = \big(a \tanh(u \cdot \bar{x})\big)_{x \in X} \in \mathbb{R}^M, \qquad h(u) = \big(\tanh(u \cdot \bar{x})\big)_{x \in X}.$$

The network output vector is the empirical average of N independent copies of Z. We prove its conditioning limit directly in this finite-dimensional space.

Lemma 3 (A smooth strictly convex rate function)**.** *Let*

$$\Lambda(\xi) = \log \mathbb{E}_{\rho_0} e^{\xi \cdot Z}, \qquad I(z) = \sup_{\xi \in \mathbb{R}^M} \{\xi \cdot z - \Lambda(\xi)\}.$$

The function Λ is finite and smooth everywhere, has positive definite Hessian everywhere, and satisfies

$$\frac{\tau^2}{2} \xi^T G \xi \le \Lambda(\xi) \le \frac{\tau^2 M}{2} \|\xi\|_2^2, \qquad G = \mathbb{E}_u[h(u)h(u)^T] \succ 0. \tag{10}$$

For every z, the supremum defining $I(z)$ has a unique maximizer ξ_z, with $\nabla \Lambda(\xi_z) = z$. The function I is finite, smooth, strictly convex, and coercive, with $\nabla I(z) = \xi_z$ and $I(z) \ge \|z\|_2^2/(2\tau^2 M)$.

Proof. Integrating out the independent Gaussian a gives

$$\Lambda(\xi) = \log \mathbb{E}_u \exp\left\{\frac{\tau^2}{2}(\xi \cdot h(u))^2\right\}.$$

Since each component of $h(u)$ lies in $[-1, 1]$, this is at most $\tau^2 \|\xi\|_1^2/2$. Jensen gives the lower bound in (10). Boundedness also permits differentiation locally to all orders under the integral.

The feature vectors $h(u)$ span $\mathbb{R}^M$. Indeed, choose a linear score taking distinct values on the M inputs. A nonzero constant tanh feature and $M - 1$ tanh features with thresholds between consecutive scores and steepness tending to infinity converge to an invertible matrix of constant and step-function columns. At some finite steepness that matrix is already invertible. Since the Gaussian u law has positive density everywhere and h is continuous, $\mathbb{E}_u(\xi \cdot h(u))^2 > 0$ for every nonzero ξ. Thus $G \succ 0$.

The law of Z has full linear span and is symmetric under $a \mapsto -a$, so it is not supported on an affine hyperplane. Its strictly positive exponential tilts have the same property. Their covariance matrices, which are the Hessians of Λ, are positive definite. The lower quadratic bound makes $\xi \cdot z - \Lambda(\xi)$ tend to minus infinity as $\|\xi\| \to \infty$. Its unique maximizer therefore exists and solves $\nabla \Lambda(\xi_z) = z$. The inverse function theorem gives

$$\nabla I(z) = \xi_z, \qquad \nabla^2 I(z) = \big(\nabla^2 \Lambda(\xi_z)\big)^{-1} \succ 0.$$

Taking the Legendre transform of the upper quadratic bound proves the asserted lower bound on I. □

For a history S, let

$$C_S = \{z \in \mathbb{R}^M : f(x)z_x \geq 1/2 \text{ for } x \in S\}. \tag{11}$$

The teacher certificate gives a strictly feasible point. The continuous coercive rate I therefore has a unique minimizer z_S on this closed convex set.

Lemma 4 (Conditional concentration)**.** *For each history S, the posterior full output vector converges in probability to z_S as $N \to \infty$. The posterior complement of any fixed neighborhood of z_S has probability decaying exponentially in N.*

Proof. Cramér's theorem applies to the empirical averages of Z, since their cumulant is finite everywhere [1, Theorem 2.2.30]. Its good rate function is I. The infimum of I over the strict version of C_S equals $I(z_S)$: mix z_S with the strict teacher mean and let the mixture weight tend to zero. Coercivity and uniqueness give a positive rate gap on the part of C_S outside any fixed open neighborhood of z_S. The large-deviation lower bound on the open conditioning event and upper bound on that closed complement prove the claim. In particular these conditioning events have positive probability for sufficiently large N. □

Define the one-neuron exponential tilt

$$\frac{d\rho_z}{d\rho_0} = \exp\{\xi_z \cdot Z - \Lambda(\xi_z)\}. \tag{12}$$

Then $\mathbb{E}_{\rho_z} Z = z$ and $\mathrm{KL}(\rho_z \| \rho_0) = I(z)$. For any law ρ with mean z and finite relative entropy, the density identity gives

$$\mathrm{KL}(\rho \| \rho_0) = \mathrm{KL}(\rho \| \rho_z) + I(z). \tag{13}$$

Finite entropy makes the required moments integrable, as (16) below also proves. Thus ρ_z is the unique minimum-entropy law of mean z, and $\rho_S := \rho_{z_S}$ is the unique entropy projection onto the training constraints. This is the usual relative-entropy, or Sanov, interpretation of the variational problem. The actual posterior convergence has already been proved directly by the finite-dimensional Cramér argument.

3 A bounded potential pays for every bad prediction

The same teacher law is feasible at every target-consistent history, so

$$0 \leq I(z_S) \leq K_T. \tag{14}$$

For the empty history, $z_\varnothing = 0$ and $I(z_\varnothing) = 0$.

Lemma 5 (Projection increment)**.** *If S' extends S by one correct training example, then*

$$I(z_{S'}) - I(z_S) \geq \mathrm{KL}(\rho_{S'} \| \rho_S). \tag{15}$$

Proof. Since $z_{S'} \in C_S$, first-order optimality for the convex minimization at z_S gives $\xi_{z_S} \cdot (z_{S'} - z_S) \geq 0$. The tilt identity gives

$$\mathrm{KL}(\rho_{S'} \| \rho_S) = I(z_{S'}) - I(z_S) - \xi_{z_S} \cdot (z_{S'} - z_S),$$

which proves the claim. □

For any ρ with $\mathrm{KL}(\rho\|\rho_0) \le K_T$, the entropy inequality and $\mathbb{E}_{\rho_0} e^{a^2/(4\tau^2)} = \sqrt{2}$ imply

$$\mathbb{E}_\rho a^2 \le 4\tau^2(K_T + c_0), \qquad c_0 = \tfrac{1}{2}\log 2. \tag{16}$$

Applying the entropy inequality to truncations of a^2 and passing to the limit also proves moment finiteness. The observable $H_x = a\tanh(u \cdot \bar{x})$ satisfies $H_x^2 \le a^2$.

For probability measures P, Q and an observable H with finite second moments, Cauchy–Schwarz gives

$$\begin{aligned} |\mathbb{E}_P H - \mathbb{E}_Q H|^2 &\le 2(\mathbb{E}_P H^2 + \mathbb{E}_Q H^2) \int (\sqrt{dP} - \sqrt{dQ})^2 \\ &\le 2(\mathbb{E}_P H^2 + \mathbb{E}_Q H^2)\,\mathrm{KL}(P\|Q). \end{aligned} \tag{17}$$

The first line factors the density difference into the sum and difference of square roots. The second uses squared Hellinger distance at most relative entropy, with squared Hellinger distance normalized as the displayed integral.

If the next query x has limiting signed output $f(x)(z_S)_x \le h$, the new training constraint forces $f(x)(z_{S'})_x \ge 1/2$. Taking $P = \rho_{S'}$, $Q = \rho_S$, and $H = f(x)H_x$ in (17), and using (14)–(16), gives

$$\mathrm{KL}(\rho_{S'}\|\rho_S) \ge \frac{(1/2 - h)^2}{16\tau^2(K_T + c_0)}. \tag{18}$$

Each such round increases the potential by this amount. Its total budget is K_T, so at most R_h rounds have $f(x)(z_S)_x \le h$, along every input order. In particular,

$$R_0 = 64\tau^2 K_T(K_T + c_0), \qquad R_{1/4} = 256\tau^2 K_T(K_T + c_0).$$

4 From the limiting predictor to the actual posterior

There are finitely many target-consistent histories: after removing repeats, each is a subset of the finite domain. For every pair (S, x) with $f(x)(z_S)_x > h$, conditional concentration gives $p_{N,S,h}(x) \to 1$. A common sufficiently large width makes all these probabilities greater than θ. Every actual confidence mistake must then be among the limiting bad rounds counted by (5). This proves the first part of Theorem 1, uniformly over input orders.

For posterior sampling take $h = 0$ and choose the common width so each limiting-good pair has wrong-sign probability at most η/M. At bad pairs use the bound one. Every order has at most $\lfloor R_0 \rfloor$ bad rounds and at most M other distinct rounds. Linearity of expectation gives (6). A union bound shows that with probability at least $1 - \eta$ no good round makes an error, giving (7). For adaptive input choices the bounds apply conditionally on the past, because the next posterior draw is fresh. Repeated observed inputs have zero error probability. This completes the proof of the theorem.

5 An explicit polynomial entropy certificate

To prove Corollary 2, let s, q be as in (8). For each teacher neuron i, take the independent Gaussian component

$$u \sim \mathcal{N}(u_i, s^2 I_D), \qquad a \sim \mathcal{N}(2Cv_i, \tau^2),$$

and let ρ_T be their uniform mixture. Its mean output is

$$\mathbb{E}_{\rho_T} H_x = 2\sum_i v_i \mathbb{E}_\zeta \tanh((u_i + \zeta) \cdot \bar{x}), \qquad \zeta \sim \mathcal{N}(0, s^2 I_D).$$

Since tanh is 1-Lipschitz and $\|\bar{x}\| = \sqrt{D}$,

$$|\mathbb{E}_{\rho_T} H_x - 2F_T(x)| \le 2\sum_i |v_i| \mathbb{E}|\zeta \cdot \bar{x}| \le 2CBs\sqrt{D} \le 1/4.$$

The signed mean is therefore strictly greater than $3/4$ on the entire cube. The Gaussian entropy formula bounds each component's entropy by

$$\frac{d\|u_i\|^2 + (2Cv_i)^2}{2\tau^2} + \frac{D}{2}(q - 1 - \log q) \le \frac{dDB^2 + 4C^2B^2}{2\tau^2} + \frac{D}{2}(q - 1 - \log q).$$

Convexity of relative entropy under mixing gives the same bound for ρ_T. No $\log C$ mixture penalty is needed.

At fixed B, τ, the first term is $O(d^2 + C^2)$, while

$$q - 1 - \log q \le -\log q = \max\left\{0, \log \frac{64C^2B^2D\tau^2}{d}\right\} = O(1 + \log C).$$

Because $d(1 + \log C) = O(d^2 + C^2)$, the certificate satisfies $K_T = O(d^2 + C^2)$ and $R_h = O(d^4 + C^4)$ at fixed B, τ, h. All constants are displayed in (5)–(9).

6 Scope and the role of the support restriction

Training defines nested convex constraints on the full mean output vector. A prediction below the test margin forces a nonzero movement before the next training margin can be met. Relative entropy pays for that movement from a bounded budget. This property does not require an IID input distribution.

If the initial prior is instead restricted to networks with $|F_N(x)| > 1/2$ at every input, the limiting feasible set becomes a union of margin orthants. It is generally nonconvex: averaging two valid outputs can create zeros at unobserved inputs. The first-order inequality in Lemma 5 need not hold in that model. We make no claim here about total mistakes for that globally restricted learner.

Two other distinctions matter. Training uses a fixed positive margin. For sign-only training at zero margin, the origin lies in the closure of every feasible set, so this rate minimizer no longer encodes the labels. Also, the confidence guarantee tests a margin strictly smaller than the training margin. The difference $1/2 - h$ pays for each error in (18).

The abstract proof extends to bounded activations whose ridge feature vectors span the finite input set, provided a finite-entropy strict teacher law exists. A Lipschitz bound also permits the Gaussian mixture certificate, with modified constants. This does not establish the same theorem for ReLU: an unbounded activation makes a neuron output a product of unbounded Gaussian quantities, and the global cumulant and entropy-to-second-moment estimates require a different argument.

References

[1] Amir Dembo and Ofer Zeitouni. *Large Deviations Techniques and Applications*, second edition. Springer, 1998.